

3D Fluid Volume Editing based on a Bidirectional Time Coupling Optimization Approach

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ABSTRACT

We propose a novel optimization approach to locally edit a 3D fluid volume. Starting from a fluid reconstruction sequence or a fluid simulation sequence, we formulate geometric deformations as a nonlinear optimization problem to match user-specified targets. To seamlessly blending the edited 3D fluid volume into an original temporal sequence, we provide a bidirectional time coupling optimization approach. This approach takes the physical properties of the previous frame and the next frame as constraints to solve the current frame and meanwhile respect the spatial-temporal consistency of editing in various scenarios. Our results indicate the intuitiveness and efficacy of our method.

Keywords: fluid editing, bidirectional time coupling, optimization approach, SPH model, 3D fluid volume

Index Terms: Computing methodologies—Modeling and simulation—Simulation types and techniques—Simulation by animation; Computing methodologies—Computer graphics—Animation—Physical simulation

1 INTRODUCTION

Generating fluid animations with desired effects is a burning question due to the prohibitively high labour required in editing fluid by explicitly tuning parameters, boundary conditions or specifying a priori a set of global fluid shape keyframes.

We have striven to address the difficulty of fluid editing. Starting from a fluid reconstruction sequence [2, 3] or a fluid simulation sequence [1, 6], we formulate geometric deformations as a nonlinear optimization problem to match user-specified targets. Unlike [4], our independent variable is the velocity field and acceleration, density, forces, and other physical properties of the fluid particles. It makes the solution to our optimization problem very flexible. To achieve seamless integration, we take the previous frame's physical properties and the next frame as constraints to constitute a bidirectional time coupling optimization approach. Without additional physical simulation, consumption [4], the current frame's edition can satisfy the spatial-temporal consistency requirement.

The key contributions of our work are as follows:

- **Seamless Integration:** Our optimization approach formulates the influence of forward-backward frames on the edited frame by extending geometric deformations with bidirectional time. It seamlessly integrates the user-edited frame into the fluid sequence to form a spatially and temporally coherent animation.

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- **Flexibility in Variables Selection:** We advocate a new deformation optimization with various physical properties selections. It perfectly broadens the optimization framework application to select physical properties to achieve desired fluid editing flexibly.

2 FLUID EDITING MODELING

This section covers the locally editing model of the 3D Fluid volume. Our input is a sequence of the restructured 3D fluid volumes [3, 5] with various known physical properties. We first briefly explain the fluid geometric deformations process, and then introduce detailedly our bidirectional time coupling optimization approach.

2.1 Geometric Deformations

We want to construct an implicit deformation function $\mathcal{D}$ to edit point $\mathcal{F}_i$ to its desired target position $\mathcal{T}_i$; namely, use $\mathcal{D} : \mathcal{F} \rightarrow \mathcal{F}^\sharp$ to represent the mapping of $\mathcal{F}_i$ with the deformed 3D fluid volume $\mathcal{F}^\sharp$. This function warps the 3D fluid volume to minimize the disparity between the target shape Eq.(1).

$$\mathcal{F}^\sharp \approx \mathcal{T} \quad (1)$$

Minimizing the Eq.(1) by setting $\Psi(\mathcal{F}^\sharp, \mathcal{T}) = 0$, we define one objective function as

$$\begin{aligned} \arg \min_{\mathcal{F}} \Xi(\mathcal{F}, \mathcal{D}, \mathcal{T}) &= \arg \min_{\mathcal{F}} \Psi(\mathcal{F}^\sharp, \mathcal{T}) \\ &= \arg \min_{\mathcal{F}} \sum_{\mathcal{F}_i \in \mathcal{F}} \Psi(\mathcal{D}(\mathcal{F}_i), \mathcal{T}) \end{aligned} \quad (2)$$

where $\Psi(\cdot, \cdot)$ is the distance field. Inspired by [4], we modify Eq. (2) to

$$\begin{aligned} \arg \min_{\mathcal{F}} \Xi(\mathcal{F}, \mathcal{D}, \mathcal{T}) \\ = \arg \min_{\mathcal{T}} \sum_{\mathcal{T}_i \in \mathcal{T}} \Psi(\text{Inte}^\alpha(\mathcal{T}_i, -A), \mathcal{F}) \end{aligned} \quad (3)$$

where A represents any physical parameter, such as the position x , or velocity v , or even the density ρ , pressure force f^{pressure} , or other physical properties. $\text{Inte}^\alpha(\cdot, A)$ denotes integration by A for a single time step Δt . In the $\text{Inte}^\alpha(\mathcal{T}, -A) = \mathcal{D}^{-1}$, $\mathcal{D}^{-1}$ is the inverse deformation function $\mathcal{D} = \text{Inte}^\alpha(\mathcal{F}, A)$. The Eq.(3) successfully transforms the independent variable from an implicit function $\mathcal{D}$ to a readily available physical quantity A .

2.2 Bidirectional Time Coupling Optimization Approach

Provided a user-selected frame K , we use Eq. (3) to match user-specified targets. Nevertheless, the deformed 3D fluid volume $\mathcal{F}^\sharp^K$ is arduous to maintain the spatial-temporal consistency with frame $K - 1$ and frame $K + 1$. The results in unexpected phenomena of fluid sequences, such as saltations, jumps, jitters, discontinuities, etc.

To integrate the edited frame into the fluid sequence to form a temporally coherent animation, we introduce a bidirectional time

coupling optimization approach, which takes the physical property A of the previous frame $K - 1$ and the next frame $K + 1$ as a constraint to solve the current frame K . According to notions of $\mathcal{D}$ and $\mathcal{D}^{-1}$, we design the bidirectional time coupling term E_{tem} to penalize the increment of fluid particles $\mathcal{F}^K$ backward and forward deformation.

$$E_{tem} = E_{tem}^- + E_{tem}^+ \quad (4)$$

where E_{tem}^- indicates that the physical quantity of the current frame is affected by the previous frame and is associated with the inverse operation $\mathcal{D}_K^{-1}$ between frame K and the target shape $\mathcal{T}$:

$$\begin{aligned} E_{tem}^- &= \sum_{\mathcal{F}_i^K \in \mathcal{F}^K} \|\mathcal{D}(\mathcal{F}_i^K) - \mathcal{F}_i^{K-1}\| \\ &= \sum_{\mathcal{T}_i \in \mathcal{T}} \|\mathcal{D}^{-1}(\mathcal{T}_i) - \mathcal{F}_i^{K-1}\| \end{aligned} \quad (5)$$

Similarly, E_{tem}^+ needs to be evaluated on the frame $K + 1$ and the target shape $\mathcal{T}$:

$$\begin{aligned} E_{tem}^+ &= \sum_{\mathcal{F}_i^K \in \mathcal{F}^K} \|\mathcal{D}(\mathcal{F}_i^K) - \mathcal{F}_i^{K+1}\| \\ &= \sum_{\mathcal{T}_i \in \mathcal{T}} \|\mathcal{D}^{-1}(\mathcal{T}_i) - \mathcal{F}_i^{K+1}\| \end{aligned} \quad (6)$$

This optimization strategy has two main dominances. We improve the robustness of the deformation of frame K significantly to eliminate the adverse effects of the single frame's inaccurate results. Moreover, the other is that we promote accuracy by using all neighbors' information (i.e., frame $K - 1$, and frame $K + 1$).

Our final objective function is

$$E_{final} = \Xi + \omega_r E_{tem} + \omega_l E_{lap} \quad (7)$$

where E_{lap} is the Laplace regularization [4]. ω_r and ω_l are the two weights. We solve the quantity A for 3D fluid volume deformation by minimizing E_{final} , and then move all fluid particles $\mathcal{F}$ along A using multiple forward and backward time steps.

3 EXPERIMENTAL RESULTS

We design the desired shape $\mathcal{T}$ by extracting ROI from arbitrary inputs, whether derived from real liquid surface acquisition, video-based reconstruction, the Eulerian or Lagrangian simulation, shallow water, or spectral methods.

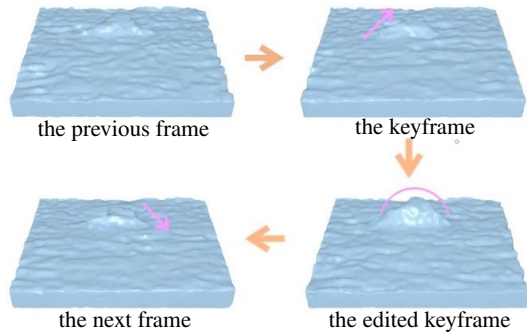


Figure 1: The higher fluid editing based on fluid's ROI.

Higher Fountain In Fig.1, we illustrate the capability of our method to edit 3D fluid volume to create a target animation seamlessly. We start from an animation that was reconstructed using the external force guiding model [3, 5]. Our edited results take the previous frame's physical properties and the next frame as constraints, thereby respecting the spatial-temporal consistency.

Multiple keyframes An exciting aspect of our method is that it enables the user to edit selected multiple keyframes locally. In Fig.2, we aim to edit the 3D Fountain volumes to generate the love-shape following user-specified four keyframes. Again, it proves that our editing results are continuous in time and space, and our method is intuitiveness and effective.

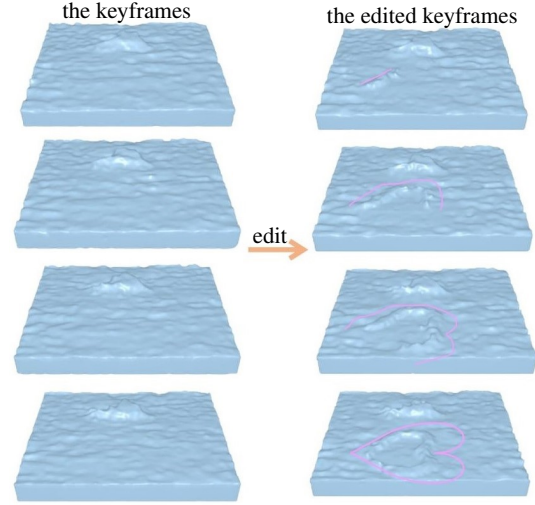


Figure 2: Fluid edit based on deformations of multiple keyframes.

4 CONCLUSION

This paper introduces the optimization approach for seamlessly editing existing fluid animations. Our method is based on a bidirectional time coupling technique, where we can consider the influence of forward-backward frames on the edited frame. This approach allows a user to form a spatially and temporally coherent animation. We can also select any physical attribute as a deformation variable, rather than just the velocity field. It allows more flexibility in the editing model.

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